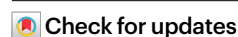


Co-firing plants with retrofitted carbon capture and storage for power-sector emissions mitigation

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Jing-Li Fan^{1,2,13}, Jingying Fu^{3,4,13}, Xian Zhang^{5,13}✉, Kai Li¹, Wenlong Zhou¹, Klaus Hubacek⁶, Johannes Urpelainen⁷, Shuo Shen^{8,1}, Shiyan Chang⁹✉, Siyue Guo⁹ & Xi Lu^{10,11,12}✉

Given that the global fleet of coal-fired power plants is mostly new, coal–biomass co-firing power plants with retrofitted carbon capture and storage (CBECCS) are regarded as a promising option for CO₂ emissions reduction. However, the effectiveness of CBECCS remains largely unexplored. Here we develop a comprehensive assessment framework featuring a macro power system combined with spatially explicit biomass sources, coal-fired units and geological storage sites. We apply this framework to investigate the spatiotemporal deployment of CBECCS in China. The results indicate that a transition to CBECCS in 2025 could supply 0.97 GtCO₂ yr^{−1} sequestration potential, with 90% at a levelized cost between \$30 and \$50 tCO₂^{−1}. A higher CO₂ mitigation of 1.6 Gtyr^{−1} could be achieved in 2040 by increasing the unit utilization hours, corresponding to a cumulative contribution of 41.2 GtCO₂ over the period 2025–2060. This study provides a useful reference for transforming coal-dominated power systems.

The world's largest coal consumer is China, whose installed coal-fired power capacity reached 1,110 GW in 2021¹, accounting for over 50% of the global total². At present, more than 80% of China's coal-fired power plants have been operational for less than 15 years³; by design, they are anticipated to continue running and lock in their associated CO₂ emissions for several decades. To avoid carbon lock-in or stranded assets^{4,5} caused by early retirement of these modern coal-fired power generation units, options to reduce CO₂ emission have been widely discussed and include switching to biomass as a fuel

or retrofitting with post-combustion carbon capture and storage (CCS) technology.

However, multiple challenges exist for fuel switching from coal to biomass for electricity generation. First, converting coal-fuelled boilers to biomass systems would require substantial technical modifications, extra investment costs and increased fuel expense due to limited biomass availability and high transportation costs^{6–8}. Second, the efficiency of a biomass-dedicated power plant is only approximately half that of coal due to the lower heating value of biomass (for example,

¹Centre for Sustainable Development and Energy Policy Research, School of Energy and Mining Engineering, China University of Mining and Technology, Beijing, China. ²State Key Laboratory of Coal Resources and Safe Mining, China University of Mining and Technology, Beijing, China. ³Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, China. ⁴College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China. ⁵The Administrative Centre for China's Agenda 21, Ministry of Science and Technology, Beijing, China. ⁶Integrated Research on Energy, Environment and Society (IREES), Energy and Sustainability Research Institute Groningen (ESRIG), University of Groningen, Groningen, the Netherlands. ⁷School of Advanced International Studies, Johns Hopkins University, NW Washington, DC, USA. ⁸BOE Technology Group Co., Ltd, Beijing, China. ⁹Institute of Energy, Environment and Economy, Tsinghua University, Beijing, China. ¹⁰State Key Joint Laboratory of Environment Simulation and Pollution Control, School of Environment, Tsinghua University, Beijing, China. ¹¹Institute for Carbon Neutrality, Tsinghua University, Beijing, China. ¹²Beijing Laboratory of Environmental Frontier Technologies, Tsinghua University, Beijing, China. ¹³These authors contributed equally: Jing-Li Fan, Jingying Fu, Xian Zhang. ✉e-mail: zhangxian_ama@163.com; changshiyan@tsinghua.edu.cn; xilu@tsinghua.edu.cn

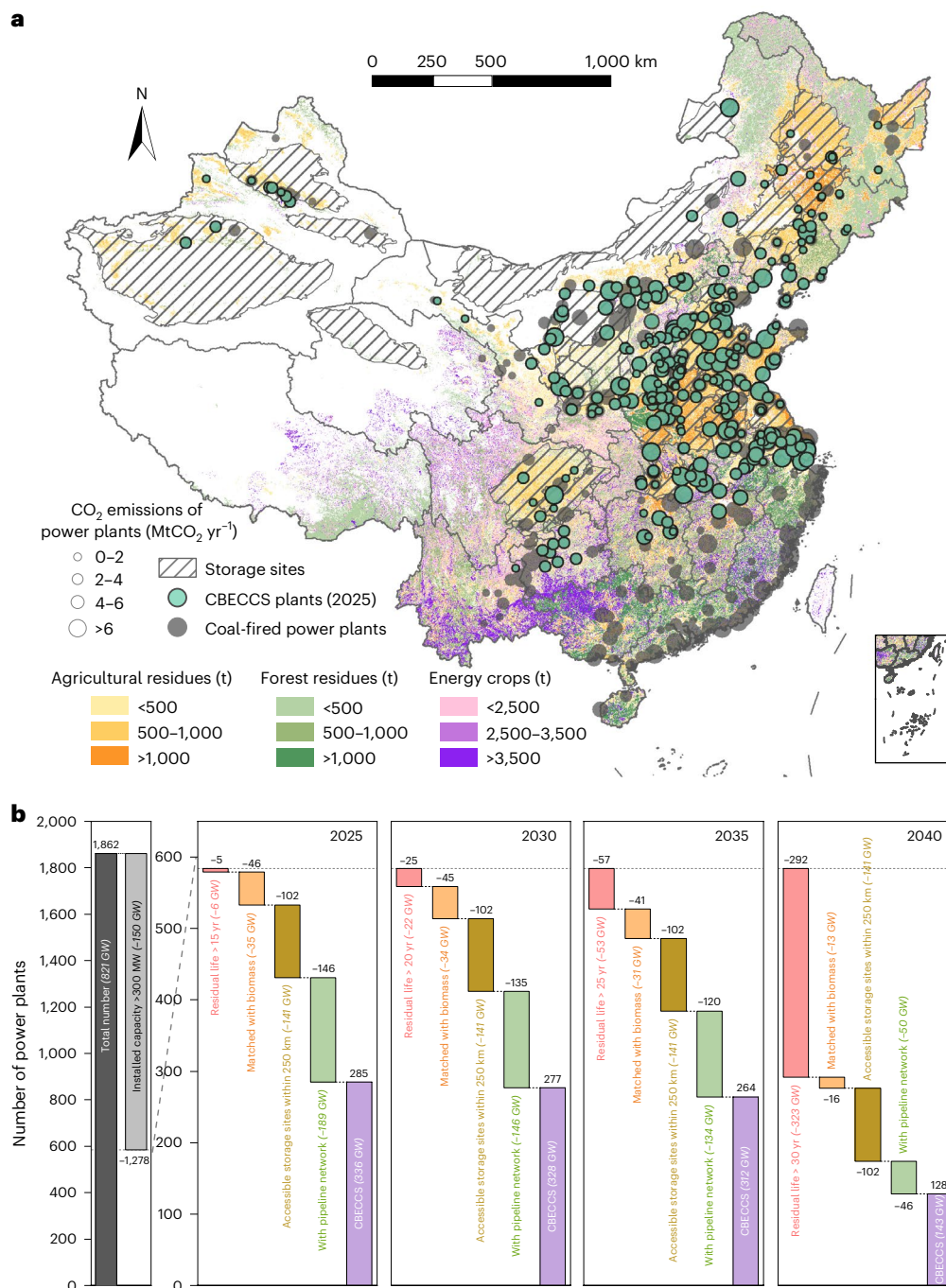


Fig. 1 | Spatial distribution of biomass, storage basins, coal-fired power plants and selection procedures of CBECCS plants in China. a, Spatial distribution of three types of biomass resources, CO₂ storage basins and selected coal-fired and CBECCS plants. The 579 selected coal-fired power plants are represented by dots, of which 285 CBECCS plants are represented by overlapping green dots. **b**, Selection procedure of CBECCS plants for China's existing coal-fired power plants under different scenarios. The grey subfigure denotes the situation of total existing power plants and those larger than 300 MW in China in 2020; the red bars show the selection procedure of meeting the minimum requirements of residual life; the orange bars show the selection procedure of

matching a certain amount of biomass within 50 km; the brown bars show the selection procedure of having accessible storage sites within 250 km; the green bars show the selection procedure of connecting to the pipeline network; the purple bars indicate the final selected CBECCS plants in China; and the four scenario years 2025, 2030, 2035 and 2040 represent the retrofit time points for all suitable CBECCS plants, after which these plants are supposed to continue to operate until the end year 2060. All provincial boundaries from the Ministry of Civil Affairs of the People's Republic of China (<http://xzqh.mca.gov.cn/map>; map content approval number: GS (2022)1873).

approximately 70% of coal⁶) and small-scale operation-induced defects (for example, lack of reheat cycles, low steam temperatures and pressures, and highly variable fuel feeds^{9,10,6,9}). Third, the widespread use of biomass power systems necessitates a significant expansion of biomass

plantations, which may compete with agricultural land^{11,12} and raise uncertainties about land-system changes and biosphere integrity^{13–16}.

Our study focuses on an alternative pathway that transforms existing coal-fired power plants to a coal–biomass co-firing system

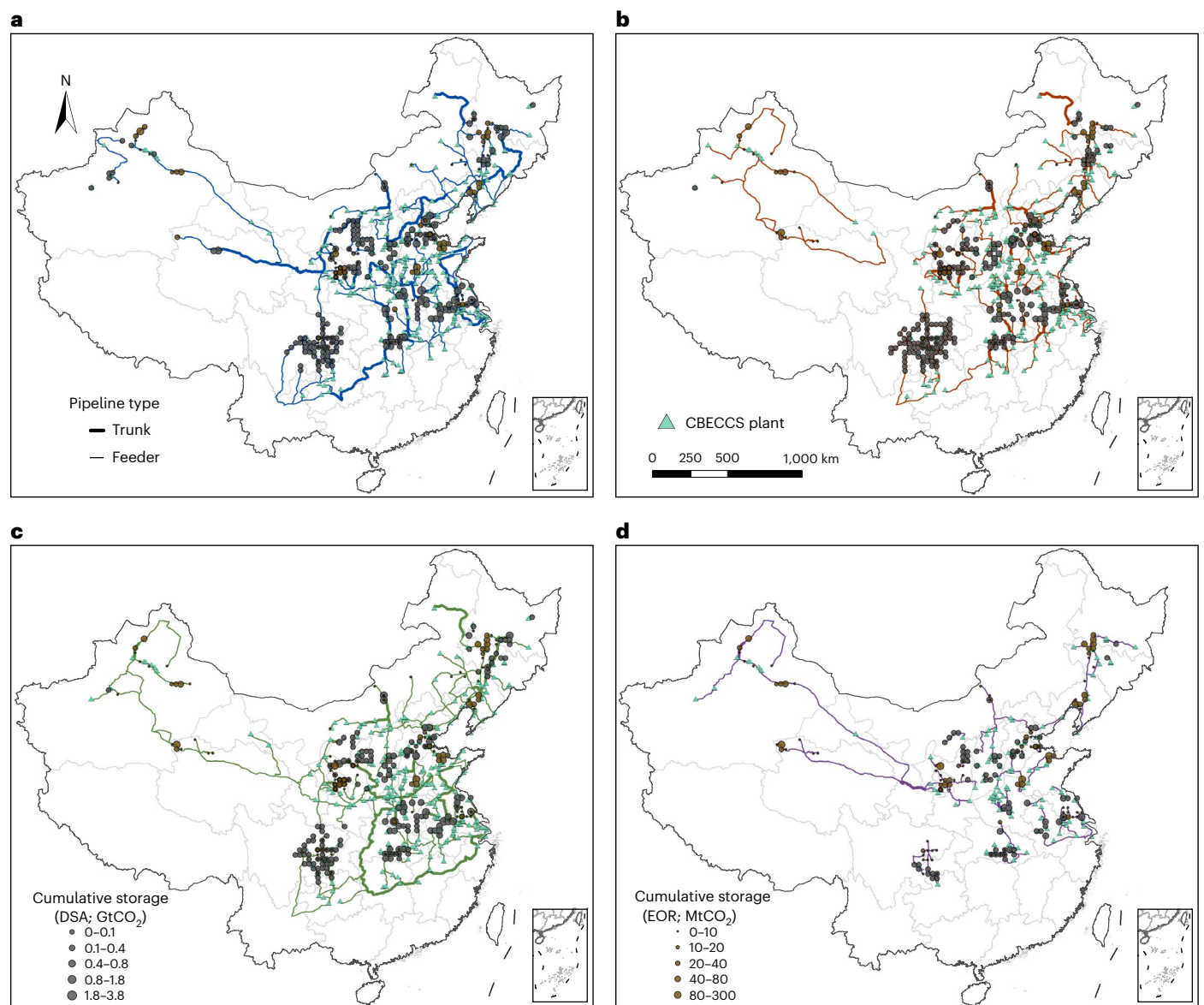


Fig. 2 | Optimal source-sink links via the pipeline network between CBECCS plants and storage sites in China under different scenarios. a, 2025. b, 2030. c, 2035. d, 2040. The green triangles represent CBECCS plants; grey circles and brown circles represent storage sites with saline aquifers and those with oilfields, respectively; circled areas of each grey or brown circle indicate CO₂ storage volume for each storage site; blue, red, green and purple lines show transportation

routes via pipelines between CBECCS plants and storage sites in four time scenarios; the thickness of the lines indicates the transported CO₂ volume through this route. DSA, deep saline aquifer. All provincial boundaries from the Ministry of Civil Affairs of the People's Republic of China (<http://xzqh.mca.gov.cn/map>; map content approval number: GS (2022)1873).

coupled with CCS (CBECCS), assuming a moderate biomass mix share, which has the following multiple advantages, especially for China: (1) relatively low boiler modification costs with a biomass share below 20–30% (refs. 17–19); (2) potential generation of carbon-neutral or even carbon-negative power if the biomass share exceeds a critical level (for example, 25% (ref. 20) based on life-cycle assessment); (3) provision of a base load for power system supplies and assurance of grid operation security in the absence of energy storage²¹; (4) economic and technical feasibility given the declining carbon capture cost (by 36% over the past 10–15 years^{22,23}) and increased successful demonstration projects in the United States, Europe and China²²; and (5) avoidance of massive asset stranding while entirely overcoming the carbon lock-in effect. The CBECCS technology is thus described and encouraged in recent Chinese initiatives towards carbon neutrality by 2060^{24–26}.

However, in contrast to the relatively large number of spatial studies on dedicated BECCS systems^{27–30}, earlier studies on CBECCS systems focused mainly on life-cycle analysis, costs or performance impacts based on specific cases, specifically in Europe and China, without considering geographical constraints^{6,20,31,32}. Several recent studies have focused on CBECCS combined with geographical heterogeneity but have investigated solely source-sink matching^{33,34} without quantifying the economic effectiveness and spatiotemporal deployment in line with a certain low-carbon emissions trajectory. Importantly, in the previous literature, CBECCS retrofitting of existing power plants has not been identified in energy system models, and such models have not been linked to a bottom-up CBECCS source-sink matching model, except for a few studies that tried to soft link a national integrated assessment model to a CCS source-sink matching model³⁵ or directly obtain exogenous emission reduction objectives³⁶.

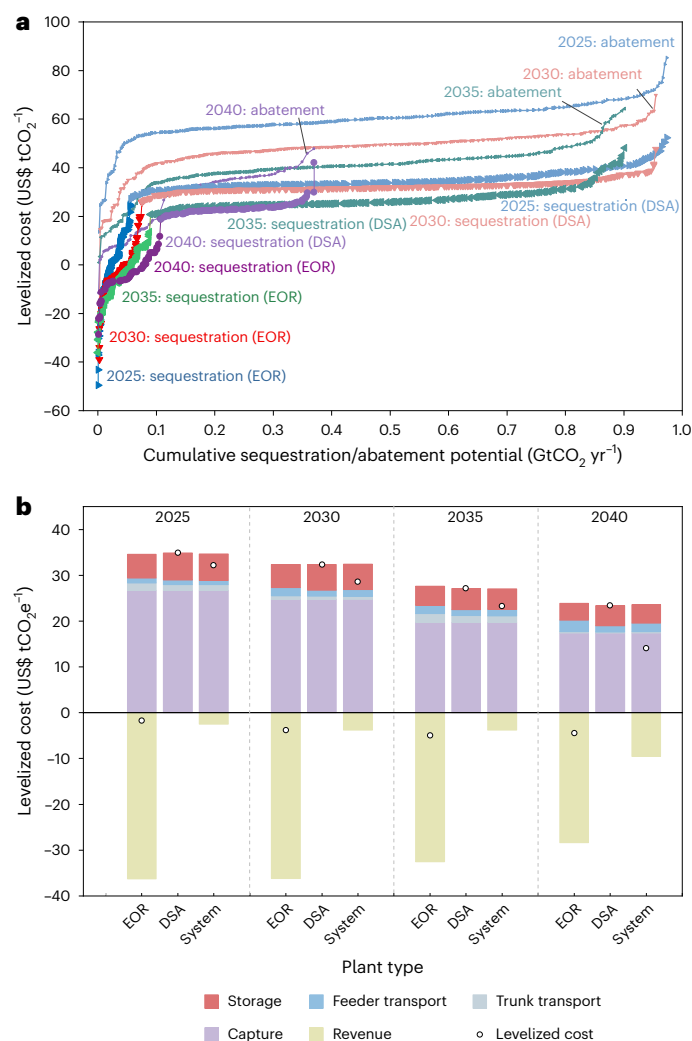


Fig. 3 | The supply potential and levelized costs under various CBECCS scenarios. **a**, The levelized cost curves describing the relationship between the levelized sequestration (abatement) costs and cumulative supply potential. **b**, Components of levelized sequestration costs for EOR-related, deep saline aquifers-related and systematic CBECCS plants. Here the levelized transportation cost is much lower than the capture cost because we adopt different data sources to keep them attached to China's cases (Methods and Supplementary Methods).

This research develops a comprehensive assessment framework to investigate the technology supply potential and CO₂ emissions mitigation contribution from CBECCS deployment as a means of retrofitting existing coal-fired power plants in China (Extended Data Fig. 1, Methods and Supplementary Methods). We first provide a single full-chain CBECCS source-sink matching optimization model for assessing the CBECCS technology supply potential under four deployment timing scenarios, which consists of four interlinked modules: biomass resource potential projection, screening of CCS retrofittable coal-fired power plants, biomass and coal-fired power plant matching and cost-minimized CO₂ source-sink matching. Second, a combination model is developed for assessing the contribution of CBECCS to carbon neutrality by 2060 in China, which dynamically soft links a full-chain CBECCS source-sink matching model and a core power system optimization model (PSOM) combined with a computable general equilibrium (CGE) model. The mapping of optimal CBECCS layout in China under two typical assessment groups and associated cost supply curves are obtained.

The following findings are highlighted. The CBECCS retrofit may have a gradually increasing contribution to mitigation for the power

system within a carbon neutrality framework, from 0.04 Gt yr⁻¹ in 2025 to a maximum of 1.58 Gt yr⁻¹ in 2040 (equivalent to 4.7% of global energy-related CO₂ emissions in 2021³⁷) and a cumulative 41.2 Gt in 2025–2060. During the time period from 2025–2040, the implementation of CBECCS results in negative CO₂ emissions of 0.01–0.25 Gt yr⁻¹ under a carbon neutrality trajectory, while net CO₂-equivalent (CO₂e) emissions are observed at 0.01–0.27 Gt yr⁻¹ if the extra life-cycle emissions are considered. The results have significant implications for decarbonizing the coal-fired power generation sector, and the methodology can also be applied to other countries.

Technology supply potential and static source-sink links

Biomass resources, CO₂ storage basins and 579 CBECCS candidate plants that can satisfy size³⁸ and age³ criteria for CCS retrofit in China show considerable geographical heterogeneities (Fig. 1 and Supplementary Figs. 1–3). Given the accessibility of available biomass, CO₂ storage sites and the potential for being along the transport pipeline network, the number of coal-fired power plants eligible for CBECCS retrofitting further narrows to 533, 431 and 285 candidate plants, respectively (Fig. 1b). These 285 CBECCS-ready power plants have a total installed capacity of 336 GW (in 746 units), representing 30.2% of total installed capacity of China's existing coal-fired power plants in 2021¹. As existing coal-fired power plants gradually approach the end of their design life (Supplementary Fig. 4), the number of CBECCS-ready power plants will fall to 277 (328 GW) by 2030, 264 (312 GW) by 2035 and sharply to 128 (143 GW) by 2040. Even in a delayed scenario, CBECCS retrofitting China's power sector will be able to meet the carbon capture retrofit demand (110 GW by 2040) in the International Energy Agency's sustainable development scenario¹⁸.

In the 2025 scenario, 295 CO₂ storage sites successfully match the CBECCS plants (Fig. 2a) via trunk (>24 inches) or feeder (<24 inches) pipelines, of which 63% (185 of 295) are situated in deep saline aquifers and 37% in oilfields for enhanced oil recovery (EOR). Unlike oilfield sites, where stored CO₂ is approaching the storage capacity limit (62–100%), saline aquifers typically have a high proportion of remaining storage capacity with an average of 24% occupation of their total storage capacity, suggesting that large-scale CBECCS deployment in China will not be limited by the storage capacity of saline aquifers but rather by that of oilfields. In addition, 29% of the total CO₂ sequestration is stored in storage sites within 50 km, primarily in the Qinshui Basin, Bohai Bay Basin and Subei Basin (Fig. 2a). Through economies of scale and shared infrastructure, these concentrated CO₂ storage sites offer prospects to construct several large-scale storage cluster centres to lower the retrofit costs¹⁸. With further phase-out of existing power plants from 2025 to 2040, the cumulative CO₂ sequestration potential by 2060 decreases dramatically from 35.1 GtCO₂ to 7.8 GtCO₂, and the overall length of the quantity-weighted transport network decreases from 48,935 km to 22,717 km (Fig. 2d).

Approximately 90% of CO₂ sequestration (0.88 of 0.97 Gt yr⁻¹) would be achieved at a cost between \$30 and \$50 tCO₂⁻¹ (Fig. 3a), which is much lower than the credits associated with the US Inflation Reduction Act (\$60 tCO₂⁻¹ for EOR and \$85 tCO₂⁻¹ for saline aquifers), despite corresponding life-cycle abatement costs reaching up to \$50–\$70 tCO₂e (Fig. 3a). Note that because of additional EOR revenues³⁹, CBECCS plants linked with oilfield sites would have profitable opportunities or lower costs (–\$50 to \$25 tCO₂⁻¹), despite having relatively long CO₂ transport distances to oilfields (185 km on average) when compared with those to saline aquifers (155 km on average). Cost cutting for the CBECCS system is greatly aided by the trunk pipeline network. For example, in the 2025 scenario, trunk pipelines have an average unit transportation cost of \$0.006 tCO₂⁻¹ km⁻¹, which is only 28% of that for feeder pipelines; however, the unit transportation cost will rise by 152% from 2025 to 2040 as the intensity of the trunk transport network layout decreases. These cost increases will

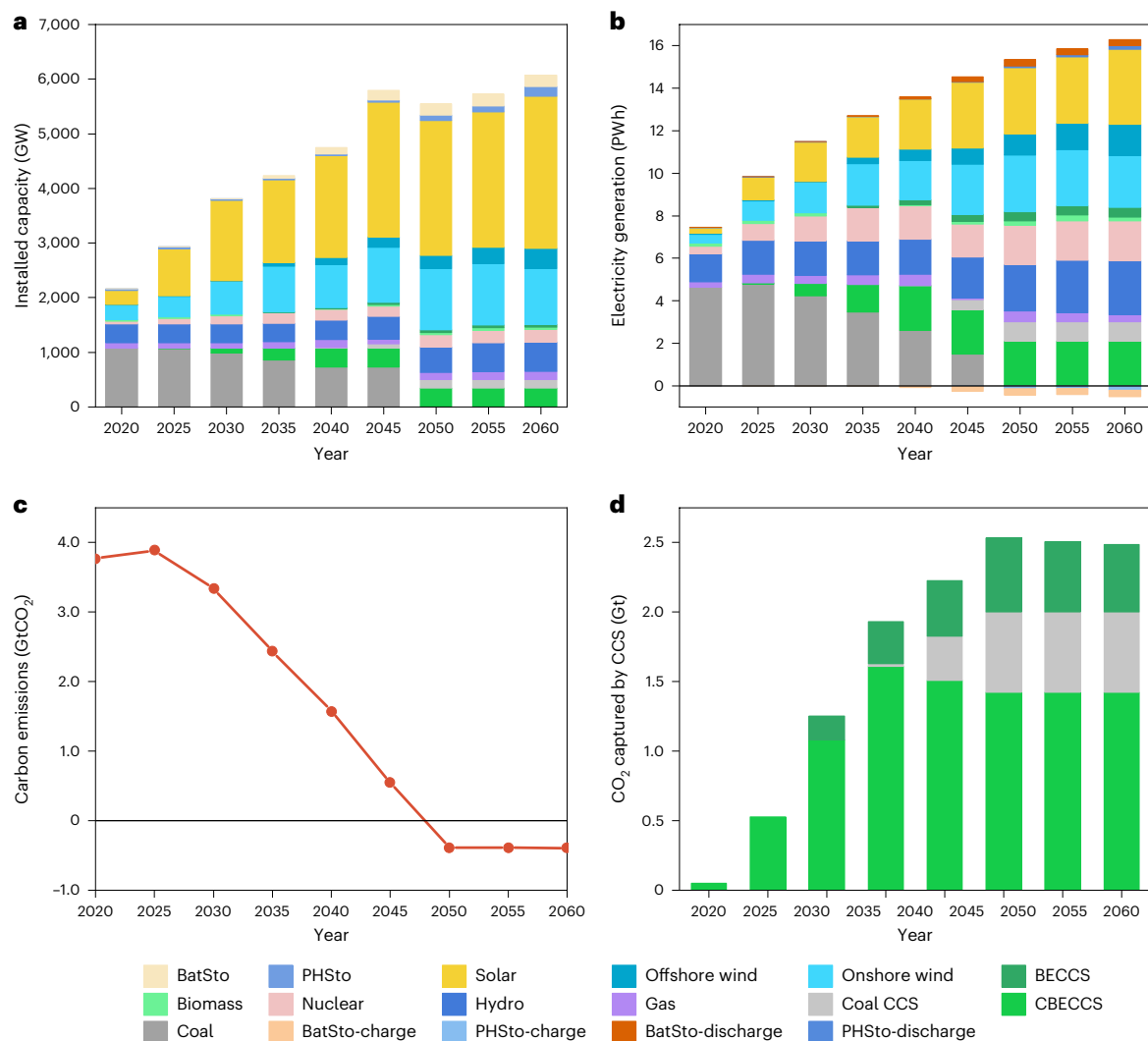


Fig. 4 | Power system model simulation results over the 2020–2060 period for scenarios including CBECCS technology with geographical constraints.

a, The installed capacity mix of China's power system. **b**, The electricity generation mix of China's power system, where the negative values indicate

power charging for energy storage. **c**, The annual carbon emissions of China's power system. **d**, The amount of CO₂ captured by CCS-related technologies. BatSto, battery storage; PHSto, pumped hydro storage.

be largely offset by the relatively high share of EOR storage and the reduced CCS-related cost associated with technology advancement, making the average CO₂ sequestration and abatement costs in the 2040 scenario just 47% and 48%, respectively, of those in the 2025 scenario (Fig. 3b).

CBECCS mitigation contribution and spatiotemporal layout

Considering the competition from other mitigation technologies (Fig. 4), CBECCS enters the power system on a small scale of 8.8 GW (1% of existing coal-fired units) in 2025, increasing to 96 GW (10%), 217 GW (25%) and the maximum retrofittable 350 GW (48%) in 2030, 2035 and 2040, respectively (Fig. 4a). This level will remain unchanged after 2040 since China's existing coal-fired power plants cannot be retrofitted due to their age. In 2060, CBECCS plants are expected to contribute 6% and 13% of China's total installed capacity and electricity generation, respectively, ranking behind solar, onshore wind and hydropower. By contrast, newly built coal-fired power plants with CCS technology will provide a substantially smaller contribution to the future power system, with total captured CO₂ accounting for 40.7% of that from CBECCS by 2060 (Fig. 4d).

In terms of intertemporal dynamic deployment (Fig. 5a), a total of 269 CBECCS plants (317 GW) are successfully matched with 211 saline aquifer storage sites in 2025–2060; 13 (10.5 GW), 64 (92.9 GW), 102 (111.4 GW) and 90 (102.2 GW) plants will be gradually deployed in 2025, 2030, 2035 and 2040, respectively. These CBECCS plants and storage sites are connected by 15,273 km of trunk pipelines and 16,167 km of feeder pipelines (Fig. 5a), resulting in a total quantity-weighted length of 31,440 km, or nearly 27% of the length of natural gas pipelines in China (116,000 km) in 2021⁴⁰, of which 930 km, 6,859 km, 13,332 km and 10,320 km of pipelines should be built in 2025, 2030, 2035 and 2040, respectively. Some of the trunk pipelines constructed in the early stage are not fully utilized by the CO₂ captured in the same stage (for example, 28% utilization rate in 2025 compared with 74% in 2040), indicating that some additional space should be reserved before the utilization time. With the aid of more accessible larger-scale shared pipeline networks, many small-scale power plants could match storage sites in subsequent phases, with an average plant-level captured CO₂ of only 5.9 Mt yr⁻¹ in 2040 compared with up to 7.7 Mt yr⁻¹ before 2030.

Economies of scale caused by a larger pipeline network, combined with technology advancements, enable significant reductions in abatement and sequestration costs at later phases (Fig. 5b). For

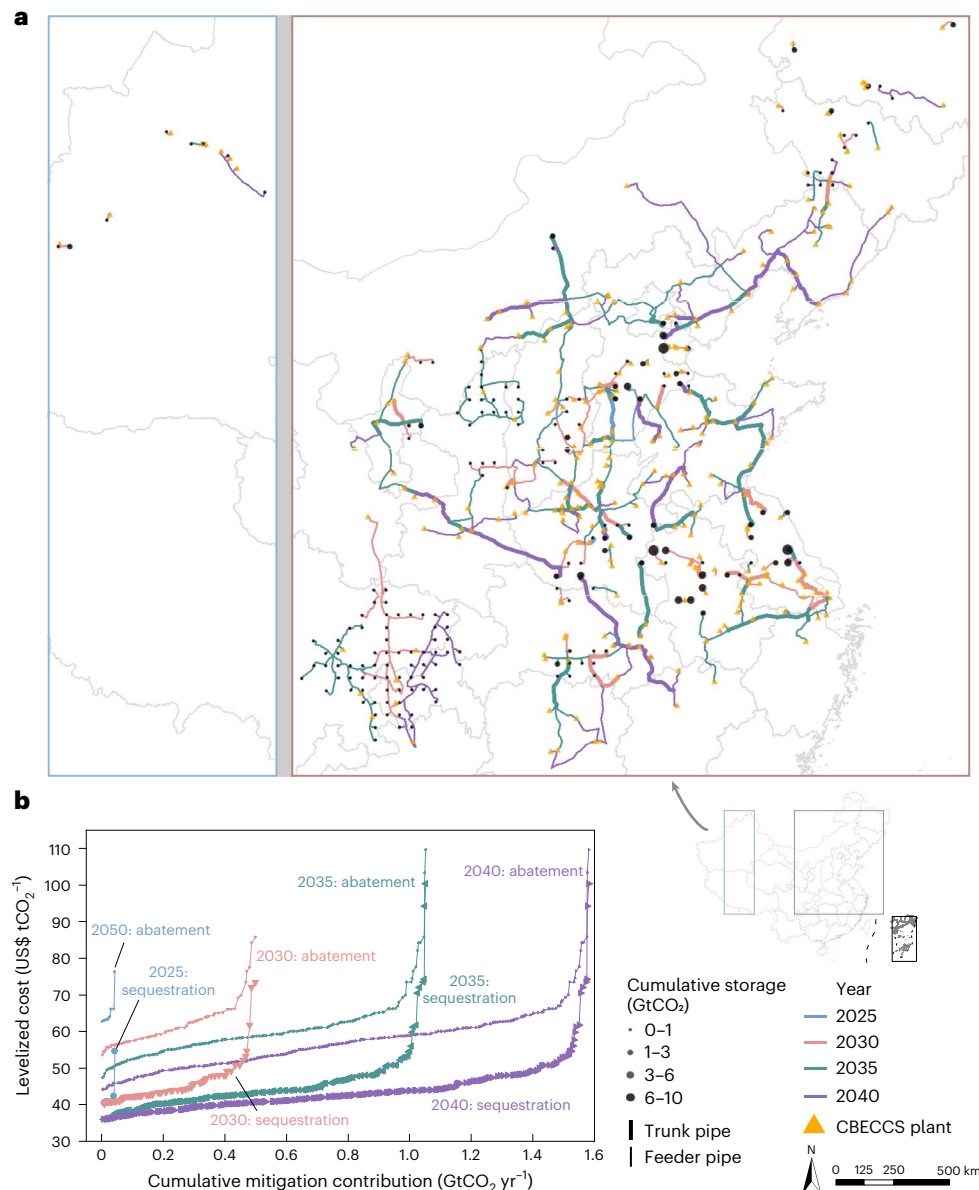


Fig. 5 | The dynamic and optimal layout for China's CBECCS deployment under various phases. a, Optimal source-sink links via the pipeline network between CBECCS plants and storage sites under various phases. Two kinds of thickness represent trunk pipelines and feeder pipelines individually. **b,** The leveled cost curves describing the relationship between the leveled

sequestration (abatement) costs and cumulative mitigation contribution. All provincial boundaries from the Ministry of Civil Affairs of the People's Republic of China (<http://xqzh.mca.gov.cn/map>; map content approval number: GS (2022)1873).

example, the median CBECCS leveled sequestration (abatement) cost would decrease from \$40.8 tCO₂⁻¹ (\$63.7 tCO₂e⁻¹) in 2025 to \$36.3 tCO₂⁻¹ (\$45.3 tCO₂e⁻¹) in 2040. For any of these phases, however, several plants would accrue very high leveled costs because they would be matched with very long feeder pipelines (for example, five plants match with a total of 2,557 km of feeder pipelines in 2040), resulting in a steep rise in the cost curve at the end of the phase. In reality, these plants can seek EOR opportunities to reduce overall sequestration (abatement) costs, similar to the results of the static model incorporating EOR. Information on trunk pipeline distribution can be combined with the low-cost range of the cost curve to identify early low-cost opportunities to develop large-scale CBECCS clusters in the Songliao Basin, Bohai Bay Basin, Junggar Basin, Ordos Basin, Subei Basin and southern North China Basin, for which the cumulative CO₂ storage is 29.8 Gt (72%) until 2060.

CO₂ emissions reduction potential from the CBECCS system

In the 2025 scenario, the supply potential of CBECCS deployment for emissions reduction would reach 1.13 Gt yr⁻¹ and cumulative 40.7 GtCO₂ by 2060, with carbon sequestration from coal combustion, bioenergy with CCS (negative emissions) and biomass substitution for coal contributing 72.6%, 13.3% and 14.2%, respectively (Fig. 6a). By subtracting the initial coal-fired emissions, net negative emissions of 55.8 Mt yr⁻¹ and cumulative 2.0 GtCO₂ by 2060 could be achieved, equivalent to nearly half of the entire CO₂ emissions of the EU-27 international aviation sector (one of the most hard-to-abate industries) in 2019⁴¹ (130 Mt yr⁻¹). After accounting for extra life-cycle emissions (127.0 MtCO₂e yr⁻¹), the net CO₂ emissions would be 70.5 MtCO₂e yr⁻¹ and cumulative 2.54 GtCO₂e. With the CBECCS retrofit delayed to the 2030–2040 period, the corresponding CO₂ emissions reduction would

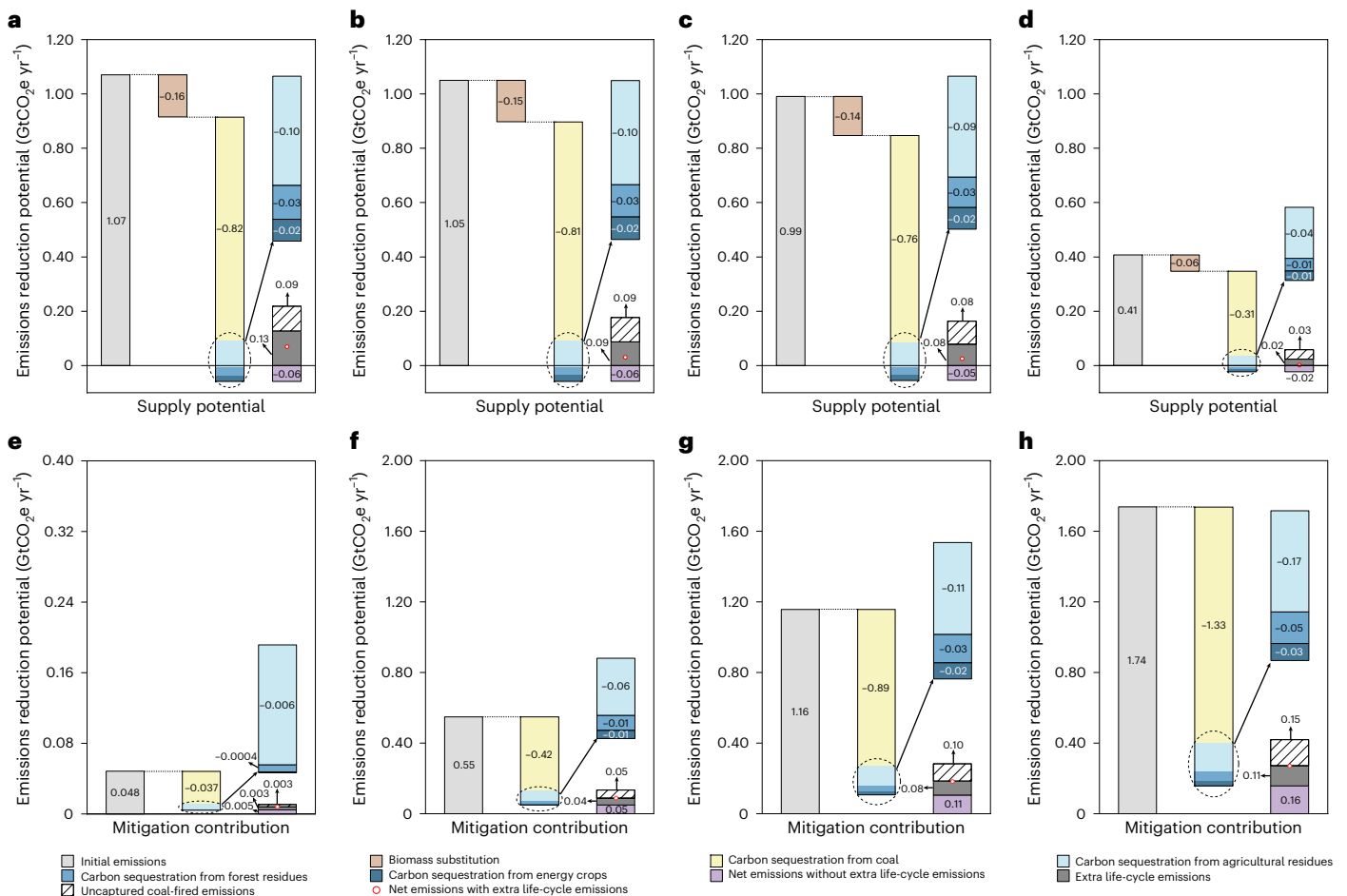


Fig. 6 | Aggregated emission reduction potential from the CBECCS system and its breakdown in the 2025–2040 scenarios. a–d, Supply potential for the years 2025 (a), 2030 (b), 2035 (c) and 2040 (d). **e–h,** Mitigation contribution for the years 2025 (e), 2030 (f), 2035 (g) and 2040 (h). The grey bars indicate the initial CO₂ emissions without CBECCS retrofit; orange bars indicate emission reduction from biomass substitution for coal, which is not reported in mitigation contribution scenarios to avoid double accounting; yellow bars indicate carbon sequestration from coal combustion; the three blue colours below the yellow bars indicate negative emissions, corresponding to the carbon sequestration

from agriculture residues, energy crops and forest residues, respectively; purple bars indicate net negative emissions calculated by subtracting total reduced emissions from initial emissions; black bars indicate extra life-cycle CO₂e emissions; and white bars indicate uncaptured CO₂ emissions, together constituting the life-cycle CO₂e emissions of the CBECCS system; the dots on top of the bars indicate the net emissions considering extra life-cycle emissions caused by CBECCS technology deployment. Here biomass substitution for coal is not considered in the mitigation contribution accounting because it was already typically considered in the power system optimization model.

drop to 0.43–1.11 GtCO₂e yr⁻¹ (including 60–145 Mt yr⁻¹ negative emissions and 20–55 Mt yr⁻¹ net negative emissions), corresponding to a cumulative 9.0–34.3 GtCO₂ by 2060 (Fig. 6b–d).

Figure 6e–h show that the CO₂ mitigation contribution (excluding biomass substitution) from the CBECCS retrofit under the carbon neutrality pathway is expected to gradually increase from 0.04 Gt in 2025 to 0.50 Gt in 2030, 1.05 Gt in 2035, and 1.58 Gt in 2040, corresponding to 26%–107% of China's projected emissions reduction requirement via CCS technology in 2050 (1.5–6.2 Gt yr⁻¹) under the 1.5 °C warming limit⁴². The negative emissions involved (246 Mt yr⁻¹) can fully satisfy the minimum mitigation requirement for bioenergy with CCS technology in 2040 (80–100 Mt yr⁻¹) towards the 2060 carbon neutrality goal according to China's carbon dioxide capture, utilization and storage annual report (2021)⁴³. Furthermore, cumulative CO₂ mitigation contribution would reach 41.2 Gt in 2025–2060, equivalent to approximately 10.3% of the 1.5 °C global carbon budget⁴⁴, although its life-cycle mitigation would decrease by 7.4%, with an annual decrease rate of 6.9–8.0%. Notably, the utilization hours endogenously observed in the PSOM are higher than those actual operation hours used for supply potential analysis. As a consequence, the mitigation contribution

through 2035 and beyond would be even greater than the estimated maximum CBECCS supply potential.

Discussions and conclusions

In this study, we developed a comprehensive assessment framework of the CBECCS system to evaluate the technology supply potential and mitigation contribution by retrofitting existing coal-fired power plants in China. The framework includes two groups of new models, a single full-chain source–sink matching model and a combination model further synthesizing the power system optimization model with a dynamic source–sink matching model. These models enable a high-resolution spatial match of adequate biomass supply, CBECCS-ready power plants and suitable CO₂ storage sites in terms of statically time scenario-divided and dynamically phased CBECCS deployment. The findings indicate that, with various deployment scenarios from 2025 to 2040, approximately 128–285 CBECCS plants with installed capacities of 143–336 GW could be matched, respectively, with 175–336 storage sites via pipeline network sizes of 22,717–48,935 km, achieving an annual CO₂ emission reduction potential of 0.43–1.13 Gt yr⁻¹. The phased CBECCS deployment in the carbon neutrality scenario

is expected to contribute progressively from 0.04 Gt yr⁻¹ in 2025 to a maximum of 1.58 Gt yr⁻¹ in 2040, with a cumulative 41.2 Gt in the 2025–2060 period via 15,273 km of trunk pipelines and 16,167 km of feeder pipelines⁴⁵. If the upper biomass mixing ratio is increased to 30% (Supplementary Fig. 5), an additional 113 Mt yr⁻¹ (12.1%) of the supply potential in the 2025 scenario and 29.9 Mt yr⁻¹ (2%) of the mitigation contribution in 2040 (and afterwards) would be achieved relative to the baseline 15%-biomass case.

Policy makers in China need to advance CBECCS deployment with spatial heterogeneity considerations (Supplementary Fig. 6). For example, the provinces of Henan (2025, optimal retrofit time), Hebei (2025), Anhui (2030) and Jiangsu (2030) in the North China Plain are ranked as top-priority candidates for large-scale CBECCS deployment with saline aquifers due to their enormous CO₂ capture and storage volumes. CBECCS combined with EOR could be preferentially deployed in Liaoning, Jilin and Xinjiang due to their highly cost-effective CO₂ sequestration and abatement (Supplementary Fig. 6). Hubei (2035), Shandong (2035), Shaanxi (2035) and Ningxia (2035) should be listed as second-priority candidates as they exhibit concentrated biomass and CO₂ emissions but insufficient sequestration capacity. Although not investigated in this study, offshore CO₂ storage could be viewed as prospective CBECCS opportunity for biomass-rich southeastern coastal regions, including Guangdong, Fujian, Guangxi and Hainan. Other provinces, for example, Gansu, Guizhou, Inner Mongolia and Chongqing, where retrofitting is extremely challenging and restricted by biomass, CO₂ emissions and storage sites, should be prioritized for phase-out.

China needs to overcome several major technical and policy challenges. First, more research-and-development investment is required to enhance the overall efficiency and cost effectiveness of the CBECCS system. Currently, very few co-firing power generation facilities are equipped with CCS^{22,43,46}, although coal and biomass co-firing technologies are maturing and being used extensively, especially in Europe^{47,48}. Co-firing technologies with relatively high biomass mixing ratios (for example, >15%) have been tested only at the laboratory or theoretical level^{49,50}. Given the energy penalty incurred during the CO₂ capture process and the use of biomass fuels (Supplementary Fig. 7), CBECCS is reported to reduce power generation efficiency by approximately 8–15%^{47,51,52}. Second, novel business models, for example, biomass collecting stations, centralized biomass pretreatment and CCS clusters with large-scale shared CO₂ pipelines or hubs, can drive down the CBECCS system cost. Third, existing successful experiences in developed countries²² can be used as references for a consistent and improved policy framework to support China's CBECCS system, which should be on the agenda, for example, incorporating CCS technology into carbon credits of the current carbon market, enacting low-carbon subsidies for CBECCS power generation, CO₂ geological storage and EOR, and concurrently raising public acceptance and awareness.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-023-01736-y>.

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Methods

Major assumptions

In this study, we assumed that large-scale deployment of coal–biomass co-firing power generation coupled with CCS technology (CBECCS) in a supply potential assessment model will occur as early as 2025, while the timing of CBECCS deployment in the mitigation contribution assessment model is endogenously determined by the model. The PSOM in the mitigation contribution assessment model introduces CBECCS technology according to the installed capacity of China's existing coal-fired power plants and the corresponding maximum CBECCS retrofit potential as well as technology parameters, and the projected CBECCS deployment scale per 5 yr step and utilization hours are endogenously determined by the model. The projected lifetime of a CCS-equipped coal-fired power plant is 40 years⁵³, and it will continue to operate until 2060 if retrofitted with CBECCS technology. Considering the biomass collection costs and potential adverse environmental impacts⁵⁴, we assumed that the maximum biomass collection radius is 50 km around a given CBECCS plant⁵⁵ and that the maximum bio-energy co-firing ratio is 15% to conduct basic analysis (30% for comparison)^{53,56,57}. We introduced the CO₂ geological storage potential and injection rate capacity (the maximum allowable CO₂ injection volume during a year based on a single well) as practical constraints in the CBECCS source–sink matching model⁵⁸. We further assumed that a CO₂ pipeline will potentially be built along existing national highways and natural gas pipelines in China to reduce the very high costs of land requisition and negotiation. Detailed information is provided in Supplementary Methods.

Overview of the CBECCS supply potential and mitigation contribution assessment framework

A comprehensive framework of full-chain CBECCS technology was developed to explore optimal CBECCS deployment from supply potential and mitigation contribution perspectives in China's power sector (Extended Data Fig. 1). The comprehensive framework includes two groups of assessment models: the CBECCS supply potential assessment model and the CBECCS mitigation contribution assessment model.

The CBECCS supply potential assessment model comprises four modules: (1) a biomass resource potential projection module to estimate three types of biomass feedstocks production (agricultural residues, forest residues and energy crops from 2025 to 2060) via a $1 \times 1 \text{ km}^2$ high-resolution geographic-information-system-based land-use model; (2) a screening module for the retrofitting of CCS-qualified coal-fired power generation units based on the installed capacity and residual lifetime; (3) an optimal matching model between biomass sources and coal-fired power plants, with constraints of a 50 km collection radius and a 15% (and 30%) biomass mix ratio; and (4) a static cost-minimized CBECCS source–sink matching model between co-firing power plants and storage sites (including deep saline aquifers and EOR, at a resolution of $40 \times 40 \text{ km}^2$), under the constraints of the storage potential, injection rate capacity, natural gas pipeline and major road networks, and upper limit of the source–sink distance (250 km).

The CBECCS mitigation contribution assessment model combines the PSOM with dynamic expansion of the source–sink matching model under the carbon neutrality goal and comprises three modules: (1) a macro top-down China dynamic CGE model that can be employed to simulate a carbon neutrality road map for the 2020–2060 period and to generate the coupling variables (electricity demand and CO₂ emission trajectory) for the power model; (2) a PSOM that can estimate long-term power capacity expansion with changes in the power technology configuration over time by linking with the CGE model. In this PSOM, we innovatively quantified the dynamically phased CBECCS retrofit capacity under the carbon neutrality goal constrained by the corresponding maximum installed capacity obtained from the source–sink matching model, unit construction and retirement, hourly electricity demand, variable renewable power supply and endogenous technology learning.

The PSOM provides the dynamic CBECCS source–sink matching model with the three variables of the total power generation, CBECCS retrofit capacity and corresponding CBECCS mitigation contribution to carbon neutrality realization at various stages; and (3) a CBECCS dynamic cost-minimized source–sink matching model that identifies those CBECCS plants with various pipeline network layouts at the different stages, thus achieving CBECCS mitigation contribution objectives by coupling with the PSOM.

Biomass resource feedstocks

Three types of biomass resource feedstocks—agricultural residues, forest residues and energy crops—were estimated by compiling $1 \times 1 \text{ km}^2$ land-use grid data for 2015 (1,298,850, 429,670 and 74,660 grids, respectively) and their future amounts from 2025 to 2060. Agricultural residues were assumed to grow at a steady rate of 1% per year over future decades according to the historical increase rate of China's grain production, while the future variation in forest residues was assumed consistent with that reported in the literature⁵⁹. Both agricultural and forest residue data were spatialized from the provincial level into $1 \times 1 \text{ km}^2$ grids according to the geographic-information-system-based land-use change projection. The total amount of energy crops (miscanthus and switchgrass⁶⁰) grown from 2025 to 2060 was then estimated by combining the distribution of suitable marginal land and the per-unit yield of energy crops. Moreover, we estimated the energy crop production for comparison using a crop growth model, Agricultural Production Systems Simulator, which considers the impacts of daily meteorological factors, soil conditions and field management on crop production. More details are provided in Supplementary Methods.

Initial CO₂ emissions

The calculation of the initial CO₂ emissions of original coal-fired power plants at the unit level is provided in Supplementary Methods.

Matching between biomass feedstocks and CCS-qualified coal-fired power plants

Referring to the CCS source–sink matching optimization model developed by Fan et al.⁶¹, a similar optimal matching model was constructed to determine the optimal links between biomass feedstocks and CCS-qualified coal-fired power plants. In this model, the matching mechanism was expressed by objective functions aimed at minimizing the total biomass transport distance equivalent while maximizing the total biomass feedstocks constrained by the biomass availability (collection radius $\leq 50 \text{ km}$ and maximum co-firing ratio $\leq 15\%$ or 30%). This model introduced two important matching priorities for power plants with higher energy consumption and biomass feedstocks located closer to power plants, as well as a one-way matching rule (the biomass feedstocks in each grid can be matched with only one power plant, while each power plant can receive biomass from several sourcing sites). Then we ran this model to generate yearly biomass feedstock matching outcomes with a maximum co-firing ratio of 15% from the CBECCS deployment year to the final year of 2060 and simulated a comparison scenario with a maximum co-firing ratio of 30%. Details on objective functions and related constraints are provided in Supplementary Methods.

CO₂ emissions before CCS deployment

The calculation of the CO₂ emissions before CCS deployment—emission reduction resulting from biomass substitution for coal, emissions originating from CBECCS plants due to coal combustion and flue gas emissions originating from CBECCS plants before CO₂ capture—are provided in Supplementary Methods.

CBECCS static source–sink matching optimization model

The CBECCS static source–sink matching optimization model was developed to obtain an overall CO₂ pipeline network layout linking all

potential CBECCS plants and storage sites (saline aquifers and oilfields) under various large-scale CBECCS deployment scenarios (2025, 2030, 2035 and 2040). Referring to recent studies^{30,35}, this static model aimed to minimize the total cost of full-chain CCS technical links, including CO₂ capture, transportation and storage, by constructing a realistic and cost-effective network-based layout, but more detailed method extension is provided in the following aspects. (1) In addition to reducing right-of-way costs by relying primarily on existing natural gas pipelines³⁰, a national-scale major road network was merged with natural gas pipelines to form a candidate CO₂ pipeline network with high geographical coverage and increased source-sink matching possibilities. (2) Oilfields were specifically identified as sequestration options considering the reality that China's current CCS demonstration projects are dominated by EOR, thus contributing to the reduction in the total cost of CBECCS deployment by accessing the network. (3) The site-scale weighted-average injection rate capacity was introduced in this model, which was determined by summing the actual injection rate capacity of each geological reservoir adjusted to the weighting factor derived from corresponding sequestration potential. The weighted-average injection rate capacity better conforms with the realistic CO₂ sequestration constraints, rather than using the maximum injection rate capacity of each reservoir⁶¹. (4) In contrast to the previous geological sequestration database⁶¹, to efficiently accommodate the relatively high CO₂ flows from trunk pipelines, China's 17 large basins were spatially processed into 40 × 40 km² grid cells (1,063 saline aquifer sites and 257 oilfield sites; Supplementary Fig. 3), and 50 injection wells were allowed at each site on the basis of engineering experience. (5) Two important prescreening processes of power plants already screened by the biomass availability were specifically designed to facilitate efficient access to candidate CO₂ pipeline networks, including the storage site accessibility (straight-line distance ≤ 250 km) and pipeline network accessibility (vertical separation ≤ 20 km). Details on the CBECCS static source-sink matching optimization model are provided in Supplementary Methods.

Overall, all CBECCS plants (those screened by the installed capacity, remaining lifetime, biomass availability, storage site accessibility and pipeline network accessibility), storage sites (those screened by the pipeline network accessibility) and candidate pipelines were compiled at the nodes and arcs as model inputs. In detail, N denotes the overall node dataset, and n was defined as any possible node (for example, CO₂ sources, CO₂ sinks and other intersections) that belongs to node dataset N . The model was adopted considering four distinct overall CO₂ pipeline network layouts under the 2025, 2030, 2035 and 2040 large-scale CBECCS deployment scenarios. The mixed-integer linear programming method was used to solve the CBECCS static source-sink matching optimization model with an objective function and related constraints as follows:

Objective function. The objective of the model was to minimize the total full-chain CCS cost attributed to all CBECCS plants, as expressed in equation (1):

$$\begin{aligned} \min \text{COST}^{\text{static}} = & \sum_i \text{Scap}_i^{\text{sta}} \times \text{Ccost}^{\text{aver}} \times S_i \\ & + \sum_j \text{Rsto}_j^{\text{sta}} \times (\text{Scost}^{\text{aver}} \times R_j - \text{EOR}^{\text{aver}} \times E_j) \\ & + \sum_{(n,m)} \sum_d (\text{Peq}_d + \text{Pom}_d) \times \text{Dist}_{(n,m)} \times P_{(n,m),d} \end{aligned} \quad (1)$$

where $\text{COST}^{\text{static}}$ is the total cost of full-chain CCS technical links attributed to all CBECCS plants; S_i is a binary variable that indicates whether power plant i can be matched with certain storage sites via pipelines (the power plant can be retrofitted as a CBECCS plant) (if this is the case, $S_i = 1$; otherwise, $S_i = 0$); $\text{Scap}_i^{\text{sta}}$ is the cumulative amount of CO₂ captured from CBECCS plant i , which equals $\text{DCE}_i^{\text{Total}}$ in Supplementary equation (15) through multiplication with a 90% capture rate; $\text{Ccost}^{\text{aver}}$

is the average capture cost per unit of CO₂ over the entire deployment period considering the technological learning rate and discount rate⁴³; R_j is a binary variable indicating whether storage site j can be linked to certain CBECCS plants via pipelines (it can be activated for CO₂ injection) (if this is the case, $R_j = 1$; otherwise, $R_j = 0$); $\text{Rsto}_j^{\text{sta}}$ is the cumulative CO₂ storage amount at storage site j until 2060; $\text{Scost}^{\text{aver}}$ is the average storage cost per unit of CO₂ over the entire deployment period considering the technological learning rate and discount rate⁴³; E_j is a binary variable that indicates whether storage site j can be identified as an oilfield site with EOR revenue (if this is the case, $E_j = 1$; otherwise, $E_j = 0$); EOR^{aver} is the average EOR revenue per unit of CO₂ over the entire deployment period considering the predicted crude oil prices by 2060, discount rate and replacement ratio of oil to CO₂ (ref. 63); $P_{(n,m),d}$ is a binary variable indicating whether arc (n,m) with diameter d should be used for CO₂ flow transportation (if this is the case, $P_{(n,m),d} = 1$; otherwise, $P_{(n,m),d} = 0$); $\text{Dist}_{(n,m)}$ is the pipeline length following the actual geographic path with arc (n,m) ; Peq_d is the initial capital cost per unit of the length for pipelines with diameter d , discounted to 2025, including primary raw material, valve station and installation costs; Pom_d is the cumulative operation and maintenance cost (4% of Peq_d in the initial deployment year) per unit of the length for pipelines with diameter d over the entire deployment period considering the discount rate; i belongs to I , which is the CO₂ source dataset covering all screened CBECCS plants; j belongs to J , which is the CO₂ sink dataset encompassing all screened storage sites; and (n,m) belongs to $\text{Arc}(n,m)$, which is the arc dataset covering all candidate pipelines, with n and m denoting the start and end nodes of a given arc, respectively. More calculation details of the CO₂ transportation cost are provided in Supplementary Methods.

Constraints. First, all CBECCS plants must achieve their CO₂ sequestration maximum, as expressed in equation (2):

$$\sum_i \text{Scap}_i^{\text{sta}} \times S_i \geq T^{\text{sta}} \quad (2)$$

where T^{sta} is the expected cumulative CO₂ sequestration that can be achieved by all CBECCS plants.

Second, CO₂ flow conservation in the CO₂ pipeline network can be determined with equation (3):

$$\begin{aligned} \text{Rsto}_n^{\text{sta}} \times R_n^N = & \text{Scap}_n^{\text{sta}} \times S_n^N + \sum_{(n,m)} \text{amount}_{(n,m)} \times \\ & \text{Period} - \sum_{(n,m)} \text{amount}_{(n,m)} \times \text{Period}, \forall n \in N \end{aligned} \quad (3)$$

where n denotes the CO₂ sources, CO₂ sinks and other intersections that belong to the overall node dataset N ; $\text{amount}_{(n,m)}$ is the CO₂ flow within arc (n,m) from nodes n to m ; while $\text{amount}_{(m,n)}$ denotes the opposite flow. If node n can be identified as a CBECCS plant, $S_n^N = 1$; otherwise, $S_n^N = 0$. If node n can be identified as a storage site, $R_n^N = 1$; otherwise, $R_n^N = 0$. Period is the entire period from the various CBECCS deployment years (2025, 2030, 2035 and 2040) to 2060.

Third, no more than one pipeline can be constructed by following the same arc, and the actual CO₂ flow along arc (n,m) cannot exceed the upper flow limit for pipelines with selected diameter d , as expressed in equations (4) and (5):

$$\sum_d P_{(n,m),d} \leq 1, \forall (n,m) \in \text{Arc}(n,m) \quad (4)$$

$$\text{amount}_{(n,m)} \leq \sum_d P_{(n,m),d} \times \text{Prate}_d, \forall (n,m) \in \text{Arc}(n,m) \quad (5)$$

where $P_{(n,m),d}$ can be obtained with equation (1) and Prate_d is the upper flow limit for pipelines with diameter d .

Fourth, the cumulative CO₂ storage at storage site j cannot exceed its storage potential, as expressed in equation (6):

$$\text{Rsto}_j^{\text{sta}} \leq \text{Csto}_j, \forall j \in J \quad (6)$$

where Csto_j is the storage potential of storage site j .

Finally, the annual CO₂ storage at storage site j cannot exceed its injection rate capacity, as expressed in equation (7):

$$\frac{\text{Rsto}_j^{\text{sta}}}{\text{Period}} \leq \text{Vin}_j \times 50, \forall j \in J \quad (7)$$

where Vin_j is the single-well injection rate capacity of storage site j , and 50 injection wells were assumed for each site on the basis of engineering experience obtained from China's EOR demonstration projects.

CBECCS extra life-cycle GHG emissions

The extra life-cycle emissions of CBECCS plants equal the sum of the GHG emissions resulting from biomass acquisition, biomass transportation, biomass pretreatment, coal mining and washing, and coal transportation, the CO₂ emissions due to the generation efficiency loss resulting from CBECCS retrofitting and the CO₂ emissions related to coal combustion released into the atmosphere in the transport process (including compression) because of leakage, extra electricity consumption for CO₂ transportation (including compression) and CO₂ storage, as expressed in equation (8) (converted into equivalent amounts of CO₂).

$$\begin{aligned} \text{CE}_{i,t}^{\text{lc}} = & \text{CE}_{i,t}^{\text{b,acq}} + \text{CE}_{i,t}^{\text{b,tra}} + \text{CE}_{i,t}^{\text{b,pre}} + \text{CE}_{i,t}^{\text{c,mw}} + \text{CE}_{i,t}^{\text{c,tra}} \\ & + \text{CE}_{i,t}^{\eta} + \text{CE}_{i,t}^{\text{CO}_2, \text{lost}} + \text{CE}_{i,t}^{\text{CO}_2, \text{trastr}} \end{aligned} \quad (8)$$

where $\text{CE}_{i,t}^{\text{lc}}$ denotes the extra life-cycle GHG emissions of CBECCS plant i in year t ; $\text{CE}_{i,t}^{\text{b,acq}}$, $\text{CE}_{i,t}^{\text{b,tra}}$ and $\text{CE}_{i,t}^{\text{b,pre}}$ are the GHG emissions resulting from biomass acquisition, biomass transportation and biomass pretreatment of CBECCS plant i in year t , respectively; $\text{CE}_{i,t}^{\text{c,mw}}$ and $\text{CE}_{i,t}^{\text{c,tra}}$ are the GHG emissions resulting from coal mining and washing and coal transportation of CBECCS plant i in year t , respectively; $\text{CE}_{i,t}^{\eta}$ denotes the CO₂ emissions due to the generation efficiency loss of CBECCS plant i in year t ; $\text{CE}_{i,t}^{\text{CO}_2, \text{lost}}$ is the leakage amount of CO₂ emissions related to coal combustion in the transport process of CBECCS plant i in year t ; and $\text{CE}_{i,t}^{\text{CO}_2, \text{trastr}}$ denotes the total CO₂ emissions due to the extra energy consumption resulting from CO₂ transportation (including compression) and CO₂ storage of CBECCS plant i in year t . Detailed calculation of each component for the CBECCS extra life-cycle GHG emissions is provided in Supplementary Methods.

Uncaptured CO₂ emissions related to coal combustion

The CBECCS plant with CO₂ capture equipment exhibits uncaptured CO₂ emissions related to coal combustion due to the 90% capture rate, with more details provided in Supplementary Methods.

CBECCS leveled cost

In this study, we measured the cost effectiveness of CBECCS technology at the plant level by using the techno-economic indicators of the leveled CO₂ sequestration cost (full-chain CO₂ capture, transportation and storage) and leveled CO₂ abatement cost based on life-cycle analysis. On this basis, the cost curves could be obtained by ranking two techno-economic indicators from lowest to highest with corresponding cumulative emissions reduction at the plant level. In addition, the leveled CO₂ sequestration cost for a given CBECCS plant can be split into two separate cost items related to saline aquifer storage and oilfield storage, respectively. Detailed calculations are provided in Supplementary Methods.

Recursive dynamic CGE model

In this study, we simulated China's future CO₂ emission trajectory in the power sector and the electricity demand under the carbon neutrality target using a China multisectoral recursive dynamic CGE model. The CGE model was constructed on the basis of Walrasian computable general equilibrium theory. Under specific technological conditions, producers employ production factors and intermediate goods inputs to produce domestic final goods, part of which enters the domestic product market as the domestic supply to meet the domestic demand while the other part is exported to foreign markets. The overall domestic demand for commodities originates from the domestic supply and imports. Consumers, such as households, enterprises and the government, earn disposable income from factor sales, tax revenues and transfer payments and allocate a portion to savings while satisfying their own consumption requirements. The savings are transformed into investments and eventually meet the domestic investment demand. Under the assumption of a perfectly competitive market, producers minimize their costs under certain production technology conditions, while consumers maximize their utility through preference selection under certain income levels, and the supply and demand for factors are optimally allocated through the factor revenue in the production process. To accurately simulate China's future electricity demand and CO₂ emissions in the power sector, we characterized the various power generation technologies in detail in the production module. Under the CO₂ emission constraint to ensure achievement of the carbon neutrality target in 2060, the CGE model was operated recursively and dynamically from 2018 to 2060 in one-year intervals. The dynamic mechanism is driven mainly by the time-series changes in the production factors, including labour-force changes and capital accumulation. Detailed information on each module of the CGE model is included in the Supplementary Methods.

Power system optimization model

In this study, we developed a high-resolution PSOM for China to simulate the installed capacity mix and power generation mix of the various power generation and electric storage technologies under the carbon neutrality target. In the PSOM, we considered 13 types of power generation technologies and 2 types of energy storage technologies (refer to Supplementary Methods for the detailed technology classification). Specifically, in addition to the power generation technologies of coal-fired power, natural gas-fired power, hydropower, nuclear power, biomass power, wind power, solar photovoltaics and newly built thermal power coupled with CCS technology, two types of CCS retrofitting technologies for coal-fired units (coal-biomass co-firing coupled with CCS technology and dedicated coal-fired coupled with CCS technology) were innovatively introduced in the model. Energy storage technologies included pumped hydro storage and battery storage. Furthermore, endogenous technology learning was considered in the PSOM model, which characterizes the dynamics of cost reduction with increasing technology deployment scale. The PSOM is a mixed-integer planning model with 5 yr planning intervals from 2020 to 2060. To solve the model in an acceptable time, we ran the model at the national level while considering the hourly wind and solar resource availabilities at the provincial level as the model constraints. And the hourly wind and solar availability factors were clustered into 25 categories together with the hourly electricity demand by the K-means method. The objective of the PSOM was to minimize the total cost, and the model constraints include the installed capacity, retrofitted capacity, operation capacity, electricity generation, carbon emissions, reserve requirements and system inertia. The objective function and core constraints of the PSOM model are introduced in this section. Detailed information on the PSOM model are included in the Supplementary Methods.

CBECCS dynamic source-sink matching optimization model

The CBECCS dynamic source-sink matching optimization model is dedicated to the design of a time-phased CO₂ pipeline network layout under

a carbon neutrality trajectory. Aiming at the overall cost minimization objective, CO₂ sequestration through time-phase CBECCS deployment by 2060 must achieve the corresponding mitigation demands derived from the PSOM. In this dynamic model, CBECCS-ready plants were identified after several screening processes, including the installed capacity, remaining lifetime, biomass availability, storage site accessibility and pipeline network accessibility. Only some CBECCS-ready plants could be identified as CBECCS plants (for retrofitting) during various dynamic phases, eventually reaching the upper limit of the permitted retrofitting capacity. In addition, CO₂ sequestration in oilfields was not considered here, in line with the PSOM. Details on the CBECCS dynamic source-sink matching optimization model are provided in Supplementary Methods.

CO₂ emission reduction through CBECCS deployment

The calculation of the total CO₂ captured through CBECCS, the total CO₂ emission reduction, the net negative emissions through CBECCS deployment and the net emissions associated with extra life-cycle emissions are provided in Supplementary Methods.

Data availability

The datasets used in this study are available online at <https://doi.org/10.17632/xrf3h89t73.2> and can also be obtained from fan@cumtb.edu.cn upon reasonable request. Source data are provided with this paper.

Code availability

The codes used in this study are available online at <https://doi.org/10.17632/tt7pc5s72k.2> and can also be obtained from fan@cumtb.edu.cn upon reasonable request.

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Author contributions

X.Z., S.C., J.-L.F. and X.L. designed the research. J.F. modelled the biomass resources. K.L., W.Z., J.-L.F. and S.S. performed the comprehensive model simulation and data compiling processes. J.-L.F. wrote the article, with major contributions provided by X.L., X.Z., K.L., W.Z., J.F., K.H., J.U., S.C. and S.G. All the authors contributed to the discussions and paper revision.

Competing interests

The authors declare no competing interests.

Additional information

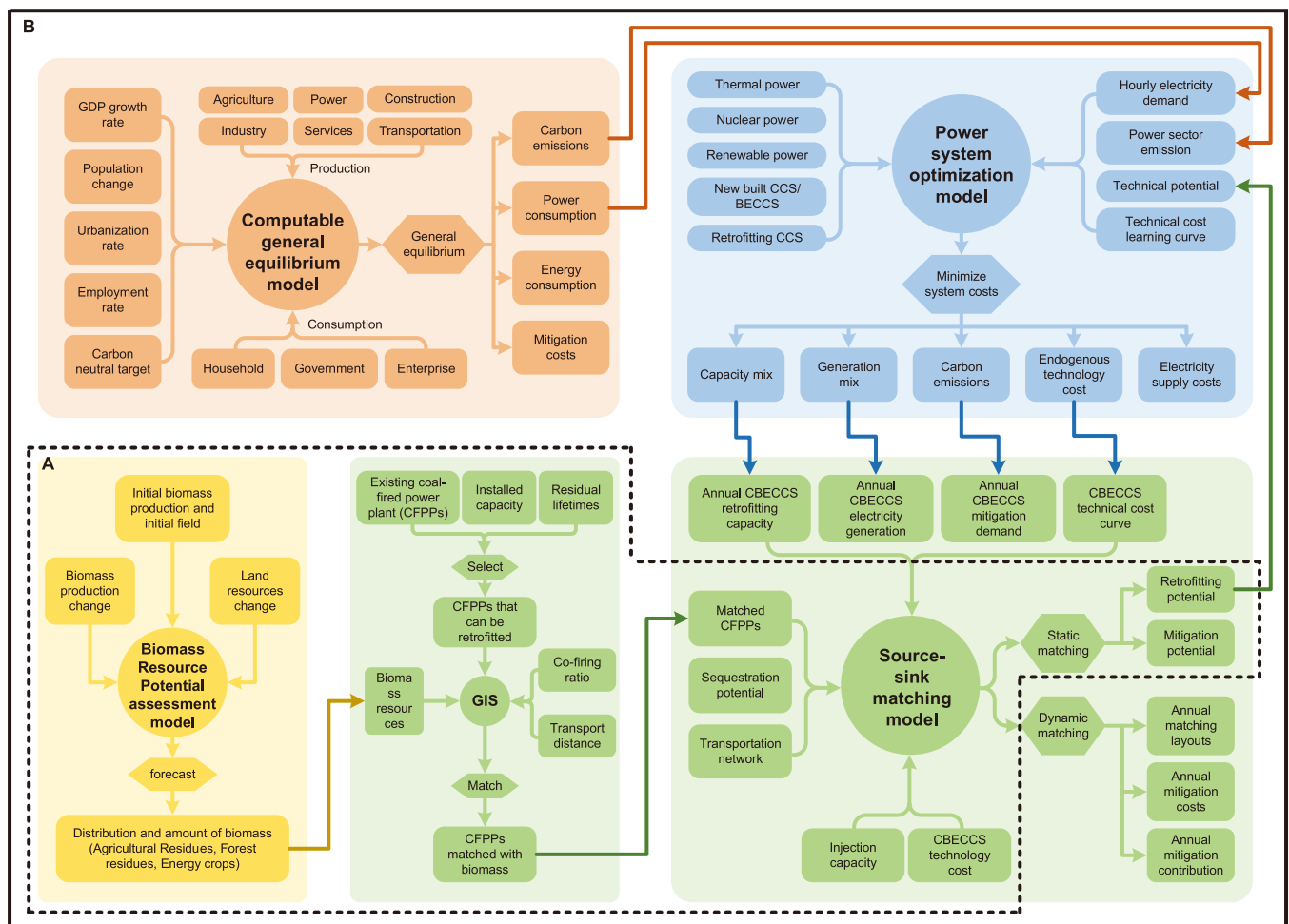
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Correspondence and requests for materials should be addressed to Xian Zhang, Shiyan Chang or Xi Lu.

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A. The CBECCS supply potential assessment model
B. The CBECCS mitigation contribution assessment model

Extended Data Fig. 1 | The comprehensive assessment model framework of full-chain CBECCS technology that consists of supply potential and mitigation contribution modules. A comprehensive framework of full-chain CBECCS technology was developed to explore optimal CBECCS deployment from supply potential and mitigation contribution perspectives in China's power sector. **A**, The CBECCS supply potential assessment model, consisting of four interlinked modules, that is, a biomass resource potential projection module, a screening module for the retrofitting of CCS-qualified coal-fired power

generation units, an optimal matching model between biomass sources and coal-fired power plants, and a static cost-minimized CBECCS source-sink matching model between co-firing power plants and storage sites. **B**, The CBECCS mitigation contribution assessment model, consisting of three interlinked modules, that is, a macro top-down China dynamic computable general equilibrium (CGE) model, a core power system optimization model (PSOM), and a CBECCS dynamic cost-minimized source-sink matching model.